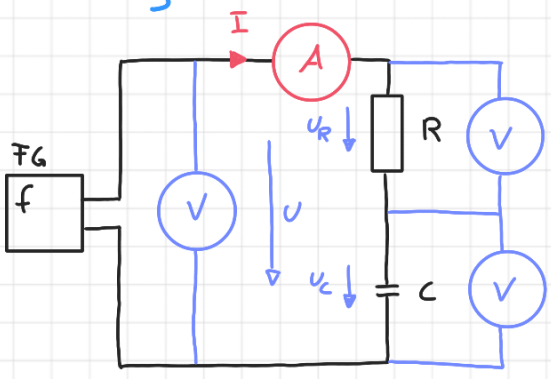
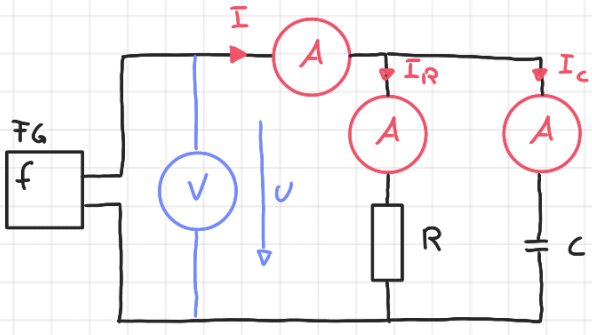
Rene Hampölz Laborübung 21.09.2021

Ortskurve

RC-Serien- & Parallelschaltung

Schaltungen





Vorgaben

$f = 4\,kHz$
$U_{eff} = 7\,V$

$I_{max} = 5\,mA$

$I_c = 50\,mA$

Dimensionierung von R

$$R = \frac{U_{eff}}{I_{max}} = \frac{7}{5 \cdot 10^{-3}} = 1{,}4\,k\Omega$$

Dimensionierung von C

$$X_c = \frac{U_{eff}}{I_c} = \frac{7}{0{,}05} = 140\,\Omega$$

$$C = \frac{1}{2\pi f \cdot X_c} = \frac{1}{2\pi \cdot 4 \cdot 10^3 \cdot 140} = 284\,nF$$

Ermitteln von U-Spitze-Spitze

$$\hat{u}_{ss} = U_{eff} \cdot 2 \cdot \sqrt{2} = 7 \cdot 2 \cdot \sqrt{2} = 19{,}8\,V$$

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Messwerte und Auswertung

Nr.	f Hz	U V	U_R V	U_C V	I mA	$X_C + R$ Ω	Realteil	Imaginärteil
1	50	6,96	0,98	6,88	0,64	$11295 \angle -83°$	1440	-11208
2	70	6,96	1,36	6,81	0,88	$8127 \angle -80°$	1440	-8006
3	100	6,96	1,90	6,65	1,24	$5776 \angle -76°$	1440	-5604
4	300	6,90	4,37	5,11	2,83	$2334 \angle -53°$	1440	-1868
5	500	6,84	5,45	3,83	3,53	$1793 \angle -39°$	1440	-1121
6	800	6,82	6,03	2,65	3,91	$1565 \angle -27°$	1440	-700
7	1500	6,81	6,38	1,50	4,13	$1449 \angle -15°$	1440	-374
8	3000	6,62	6,35	0,75	4,22	$1412 \angle -8°$	1440	-187
9	4000	6,32	6,11	0,54	4,23	$1407 \angle -6°$	1440	-140

Nr.	f Hz	U V	I_R mA	I_C mA	I mA	$X_C \parallel R$ Ω	Realteil	Imaginärteil
1	50	6,97	23	16	4,44	$1389 \angle -7°$	1378	-172
2	70	6,76	23	16	4,48	$1379 \angle -10°$	1358	-238
3	100	6,76	23	16	4,57	$1358 \angle -14°$	1318	-329
4	300	6,76	23	16	5,80	$1120 \angle -37°$	896	-672
5	500	6,75	23	16	7,68	$875 \angle -51°$	547	-683
6	800	6,73	23	18	10,97	$626 \angle -63°$	280	-560
7	1500	6,61	23	24	19,12	$361 \angle -75°$	93	-349
8	3000	6,10	23	37	36,18	$185 \angle -82°$	24	-184
9	4000	5,60	23	47	46,65	$139 \angle -84°$	14	-139

→ Berechnung von $X_C + R$ (Z) (Bei f = 500 Hz)

$$X_C = \frac{1}{2\pi f C j} = \frac{1}{2\pi \cdot 500 \cdot 284 \cdot 10^{-4} j} = 1120 \angle -90° \, \Omega$$

$$Z = R + X_C = 1400 + 1120 \angle -90° = 1793 \angle -39° \, \Omega$$

→ Berechnung von $X_C \parallel R$ (Z) (Bei f = 500 Hz)

$$X_C = \frac{1}{2\pi f C j} = \frac{1}{2\pi \cdot 500 \cdot 284 \cdot 10^{-4} j} = 1120 \angle -90° \, \Omega$$

$$Z = \frac{1}{\frac{1}{R} + \frac{1}{X_C}} = \frac{1}{\frac{1}{1400} + \frac{1}{1120 \angle -90°}} = 874 \angle -51° \, \Omega \rightarrow 546 - 682 j$$

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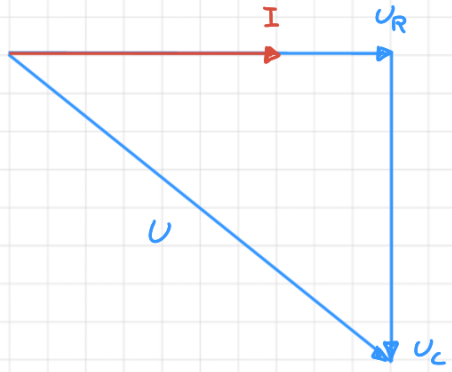
→ Zeigerdiagramm Serienschaltung (Bei f = 500 Hz)

$$U = Z \cdot I = 1793\angle{-39^\circ} \cdot 3{,}53 \cdot 10^{-3} = 6{,}33\angle{-39^\circ}\,V$$

$$U_R = R \cdot I = 1400 \cdot 3{,}53 \cdot 10^{-3} = 4{,}95\,V$$

$$U_C = X_C \cdot I = 1120\angle{-90^\circ} \cdot 3{,}53 \cdot 10^{-3} = 3{,}95\angle{-90^\circ}\,V$$



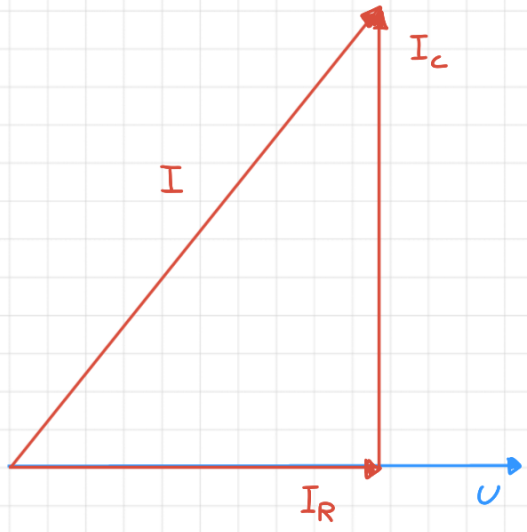
1 V ≙ 1 cm
1 mA ≙ 1 cm

→ Zeigerdiagramm Parallelschaltung (Bei f = 500 Hz)

$$I = \frac{U}{Z} = \frac{6{,}75}{874\angle{-51^\circ}} = 7{,}72\angle{51^\circ}\,mA$$

$$I_R = \frac{U}{R} = \frac{6{,}75}{1400} = 4{,}82\,mA$$

$$I_C = \frac{U}{X_C} = \frac{6{,}75}{1120\angle{-90^\circ}} = 6{,}03\angle{90^\circ}\,mA$$



1 V = 1 cm
1 mA = 1 cm

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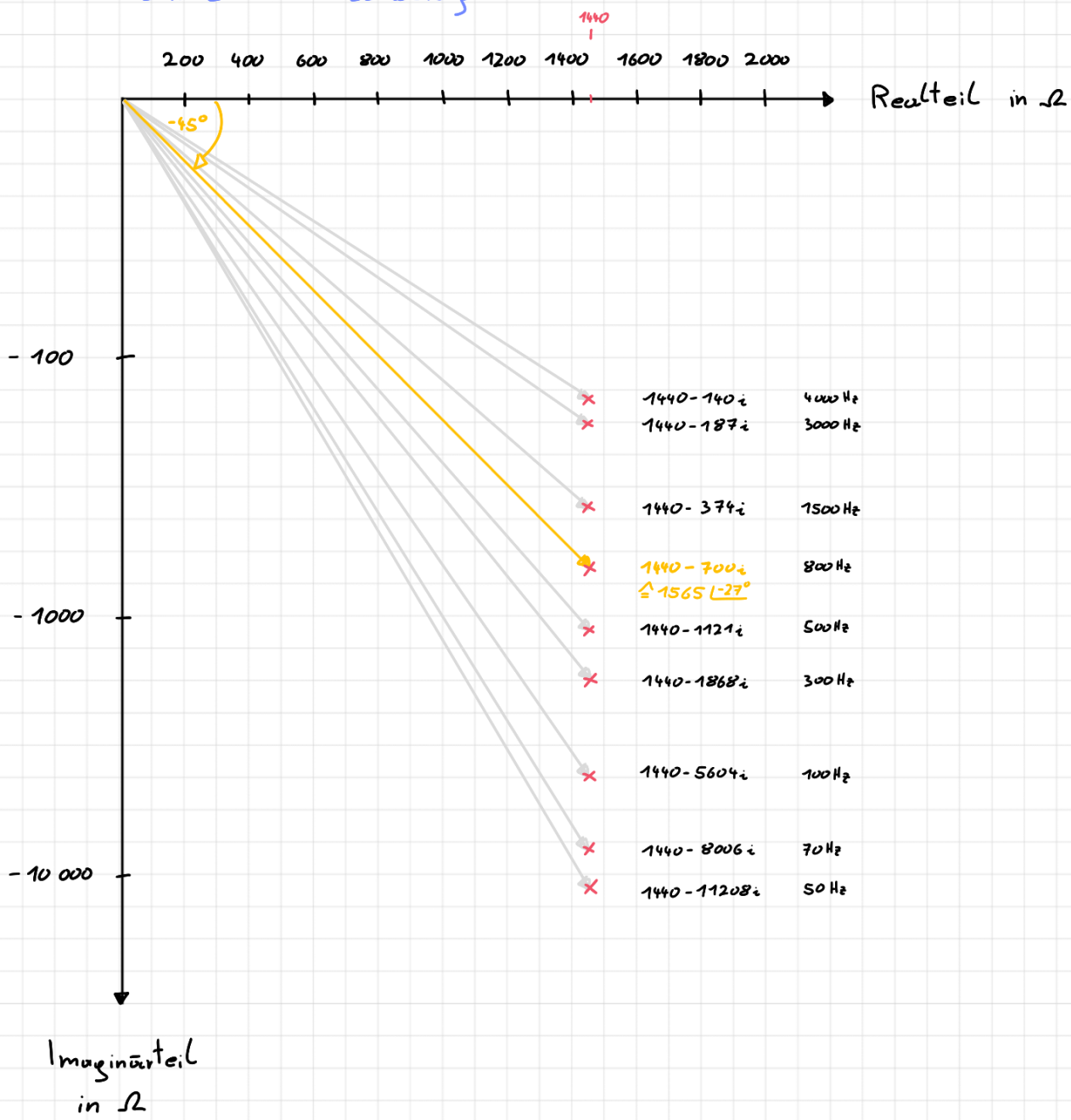
→ Ortskurve Serienschaltung



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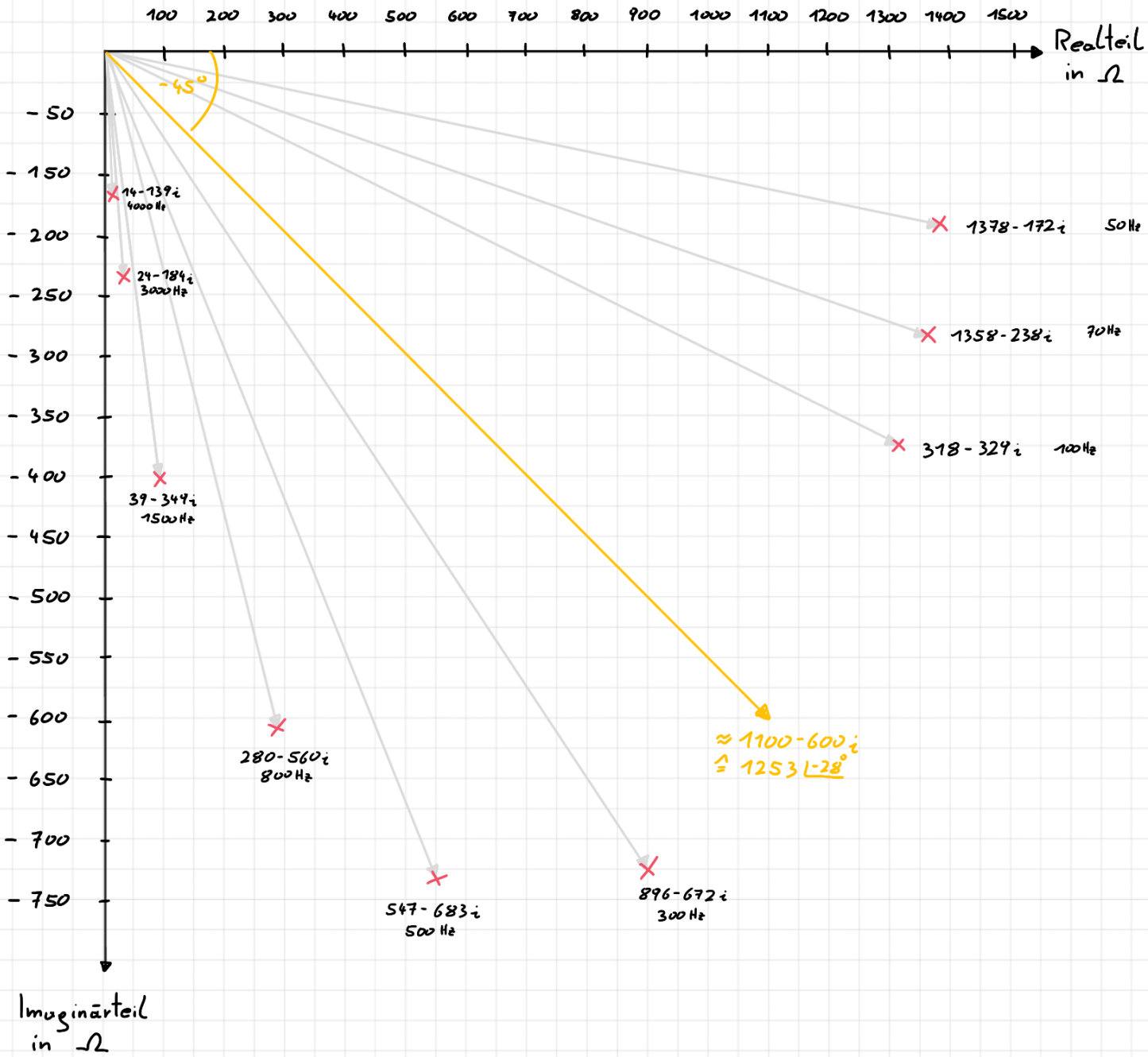
→ Ortskurve Parallelschaltung



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Verwendete Geräte

Frequenzgenerator	ET-MTL1-FG02	
Multimeter	ET-MTL1-DM22	I
	ET-MTL1-DM08	U_R, I_R
	ET-MTL1-———	I_C
Voltmeter	ET-MTL1-DM14	U
	ET-MTL1-DM12	U_C
Widerstandsdekade	ET-MTL1-RD26	R
Kondensatordekade	ET-MTL1-CD13	C

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